Finding a River Plume from Space, and Clearing Up Murky Water

Sentinel-2 plume mapping and underwater image enhancement in Python

Davina Whata, Michigan Technological University, September 2026

1. Sentinel-2 Imagery from the Copernicus Data Space

A Copernicus Data Space account was set up, and OAuth credentials were configured to access the Sentinel Hub Process API through the sentinelhub Python package. This was used to request and download Sentinel-2 imagery for two locations, shown below.

Ontonagon River mouth: true color image

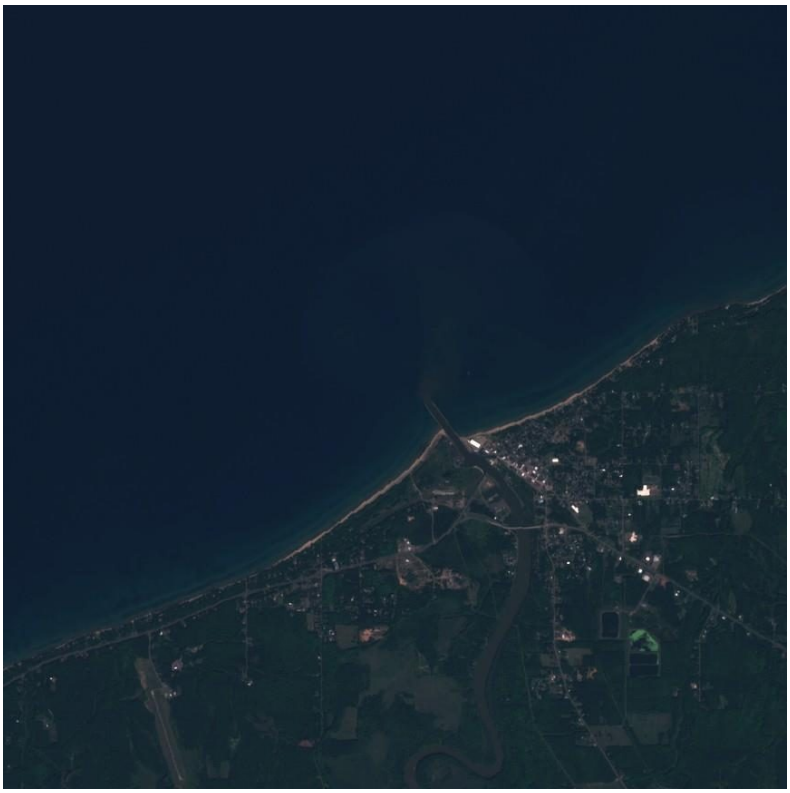


Figure 1. Sentinel-2 true color image of the Ontonagon River mouth on Lake Superior, Michigan.

Parameter	Value
Location	Ontonagon River mouth, MI (approx. 46.88044 N, -89.33164 W)
Date acquired	September 2, 2026
Bands used	B04 (Red), B03 (Green), B02 (Blue): true color composite
Processing level	Sentinel-2 L1C (top-of-atmosphere reflectance)
Requested resolution	10 m / pixel
Image dimensions	865 x 864 pixels

The river mouth itself is visible as the narrow channel where the Ontonagon River meets Lake Superior, running from the bottom of the image toward the shoreline near the center. Extending out from this point into the lake is a lighter, tan-grey plume of water that contrasts with the darker blue of the surrounding open lake. This plume is sediment- and organic-matter-laden river water dispersing into the lake, and it fans out roughly northwest of the river mouth in this image. This date was selected because it was one of the few cloud-free scenes in which the plume was clearly visible; several other dates considered had either heavy cloud cover or showed little to no visible plume.

Lake Erie shoreline: false color image (non-RGB bands)



Figure 2. Sentinel-2 false color image of the Lake Erie shoreline near the Ohio coast.

Parameter	Value
Location	Lake Erie shoreline, OH (near Toledo/Maumee Bay)
Date acquired	August 28, 2026
Bands used	B08 (NIR), B04 (Red), B03 (Green): false color composite
Processing level	Sentinel-2 L1C (top-of-atmosphere reflectance)
Requested resolution	10 m / pixel
Image dimensions	1189 x 857 pixels

Western Lake Erie was chosen for the second image because it is known for its harmful algal blooms, making it a useful real-world case for testing a band combination other than RGB. This composite was built by moving the near-infrared band (B08) into the red channel, the red band (B04) into the green channel, and the green band (B03) into the blue channel, a combination sometimes referred to as a "color-infrared" composite. Healthy vegetation strongly reflects near-infrared light, so land cover such as farm fields and shoreline vegetation appears bright red, while open water, which absorbs NIR, stays dark. A faint lighter streak is visible in the water near the shoreline bend, consistent with a low-level sediment

or algae signature, though it is not a strong, obvious bloom on this particular date.

Other Sentinel-2 bands and their uses

Beyond the standard RGB bands, Sentinel-2 carries several other bands that are useful depending on the research question. **B08 (NIR, 842 nm)** is strongly reflected by healthy vegetation and absorbed by open water, making it useful for distinguishing land/water boundaries and gauging vegetation vigor. **B11 / B12 (SWIR)** are sensitive to moisture content, useful for detecting burn scars, drought stress, and telling snow and clouds apart. **B05, B06, B07 (Red Edge)** are sensitive to chlorophyll content, useful for detecting algal blooms and monitoring crop health. **B01 (Coastal aerosol)** is primarily used for atmospheric correction, but is also helpful for picking up fine suspended sediment or aerosols close to shorelines. For research involving water quality or river plumes specifically, the Red Edge and NIR bands would likely be the most useful going forward, since they are more directly tied to chlorophyll and suspended sediment concentration than the visible RGB bands alone.

2. Mapping the Plume

A binary plume mask was built from the true color Ontonagon River mouth image (Figure 1) using a brightness- and color-based rule. Sample regions were first inspected to compare pixel statistics between clear open water, the visible plume, and the generic nearshore fringe.

Region	R / G / B (mean)	Blue fraction	Brightness
Clear water	15.0 / 27.4 / 47.2	0.53	29.8
Plume	15.9 / 32.1 / 49.7	0.51	32.6
Nearshore fringe	14.9 / 34.1 / 50.1	0.51	33.1

These statistics showed that brightness alone gives only a small separation between the plume (32.6) and the generic nearshore fringe (33.1), almost the same value. A global brightness threshold therefore could not distinguish the plume from ordinary shallow water found along the entire coastline, and an initial attempt at a mask confirmed this by lighting up the full shoreline rather than just the plume near the river mouth.

To address this, the following rule was used: water was first separated from land using a blue-fraction threshold (blue_fraction > 0.47), and candidate plume pixels were then selected as water pixels above a brightness threshold (brightness > 31.5), restricted to a region of interest (ROI) around the river mouth. A shoreline buffer (18-pixel dilation of the land mask) was subtracted to exclude water immediately adjacent to land, since this narrow strip has a similar brightness signature to the plume itself but sits much closer to shore. The resulting mask was then cleaned using morphological dilation, erosion, and dilation (disk radii 3, 5, and 3 respectively) to close small gaps and smooth the boundary, followed by a largest-connected-component filter to remove any remaining isolated speckle noise.

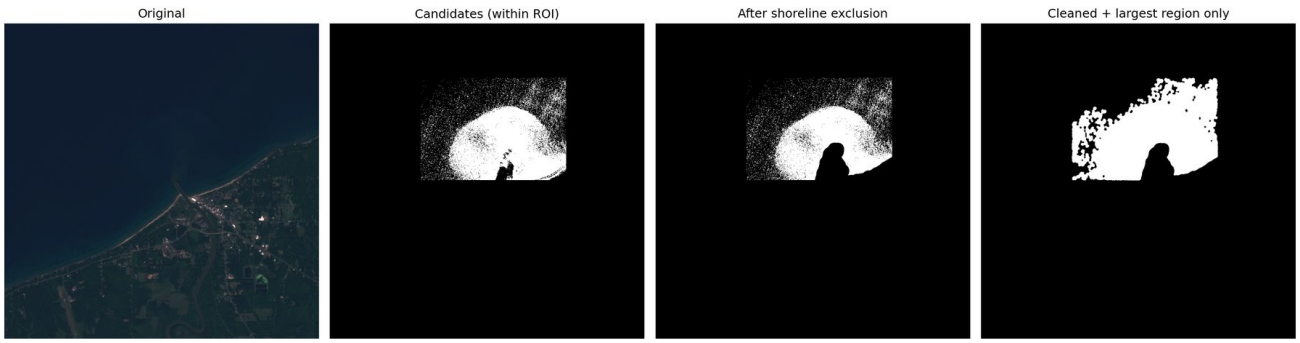


Figure 3. Progression of the plume mask: original image, brightness/ROI candidates, mask after shoreline exclusion, and the final cleaned mask (largest connected region only).

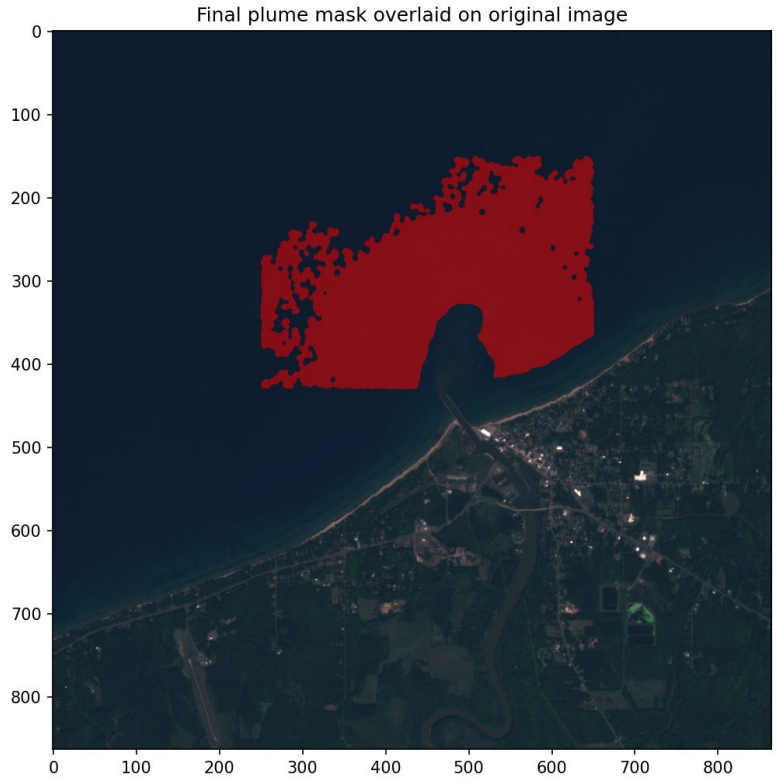


Figure 4. Final binary plume mask (red) overlaid on the original true color image. The mask closely follows the visible plume swirl and correctly excludes the small point of land within the search region.

Simple improvements to the method

A few refinements would help make this more robust: restricting the plume search to a region of interest near the river mouth (already applied here, since a brightness-only rule otherwise picks up shallow water along the entire coastline, not just the plume); using a largest-connected-component filter to discard scattered speckle noise instead of relying on erosion radius alone, which risks eating into the true plume shape; and using a ratio such as blue-fraction relative to a local (rather than global) water baseline, since lighting and water color can vary across a large scene.

Using chlorophyll- and moisture-sensitive bands

Instead of relying on visible RGB brightness alone, the same threshold-and-mask approach could be applied to Sentinel-2's Red Edge bands (B05-B07), which are sensitive to chlorophyll

absorption/reflectance and would more directly flag algae or phytoplankton concentrations tied to the plume, rather than just sediment-driven brightness changes. Similarly, NIR (B08) or SWIR (B11/B12) bands respond to water turbidity and moisture content, and could be combined with the visible bands (e.g., an NDVI-like ratio between NIR and Red) to build a rule that separates plume water from clear water based on a physical property (chlorophyll or sediment concentration) instead of a visual appearance that can vary with lighting, angle, or scene brightness.

Rule-based masking vs. deep learning segmentation

Advantages: This rule-based approach is fast, fully interpretable (every threshold and step has a clear physical justification), requires no training data or labeled examples, and can be built and adjusted directly from a handful of sample pixels. It also runs efficiently with minimal computational resources.

Disadvantages: It is brittle. Thresholds tuned to this scene (this date, this lighting, this location) may not transfer well to a different image, season, or water body, as seen when the same brightness threshold that worked for the river mouth also picked up the entire coastline before the region-of-interest restriction was added. A deep learning segmentation model, once trained on a large and varied dataset, could generalize much better across different lighting conditions, cloud cover, and plume shapes, and could learn more complex, nonlinear relationships between bands than a simple threshold rule can capture. The tradeoff is that it requires substantial labeled training data, more computational cost to train, and is far less interpretable: it is harder to explain exactly why the model classified a given pixel as plume or not.

3. Enhancing Underwater Images

Four underwater images were analyzed and enhanced using a combination of histogram analysis, gamma correction, histogram equalization, and noise reduction / edge enhancement filters.

RGB channel histograms

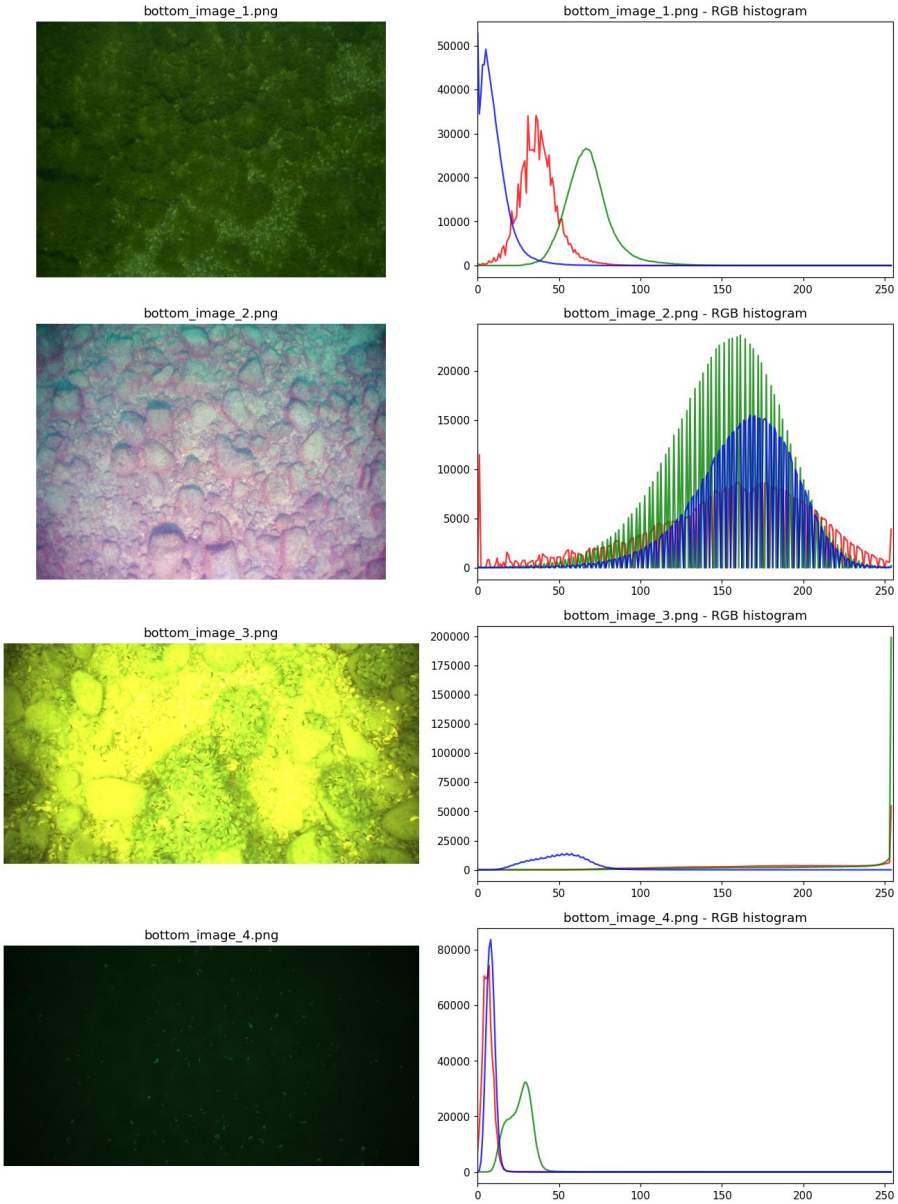


Figure 5. Original images and their RGB channel histograms.

Image	Mean R	Mean G	Mean B	Interpretation
1	37.3	68.1	10.1	Blue crushed near 0; green cast; underexposed
2	153.0	152.2	163.9	Balanced, mid-range; only a mild cast
3	187.6	214.1	50.5	Red/green clipped near 255; overexposed, yellow-green cast
4	6.3	25.8	8.0	All channels near 0; severely underexposed

None of the four images use the full 0-255 range in any channel: they are either crushed toward black (Images 1 and 4), clipped toward white (Image 3), or simply narrow (Image 2). This is the underlying problem that gamma correction and histogram equalization address: each channel's narrow range can be stretched back out to recover contrast and, in some cases, correct part of the color cast. Based on

these histograms, the blue channel needed the most correction in Images 1 and 4 (it was almost entirely crushed near zero), while the red and green channels needed correction in Image 3 (both clipped at the top of the range).

Gamma correction vs. histogram equalization

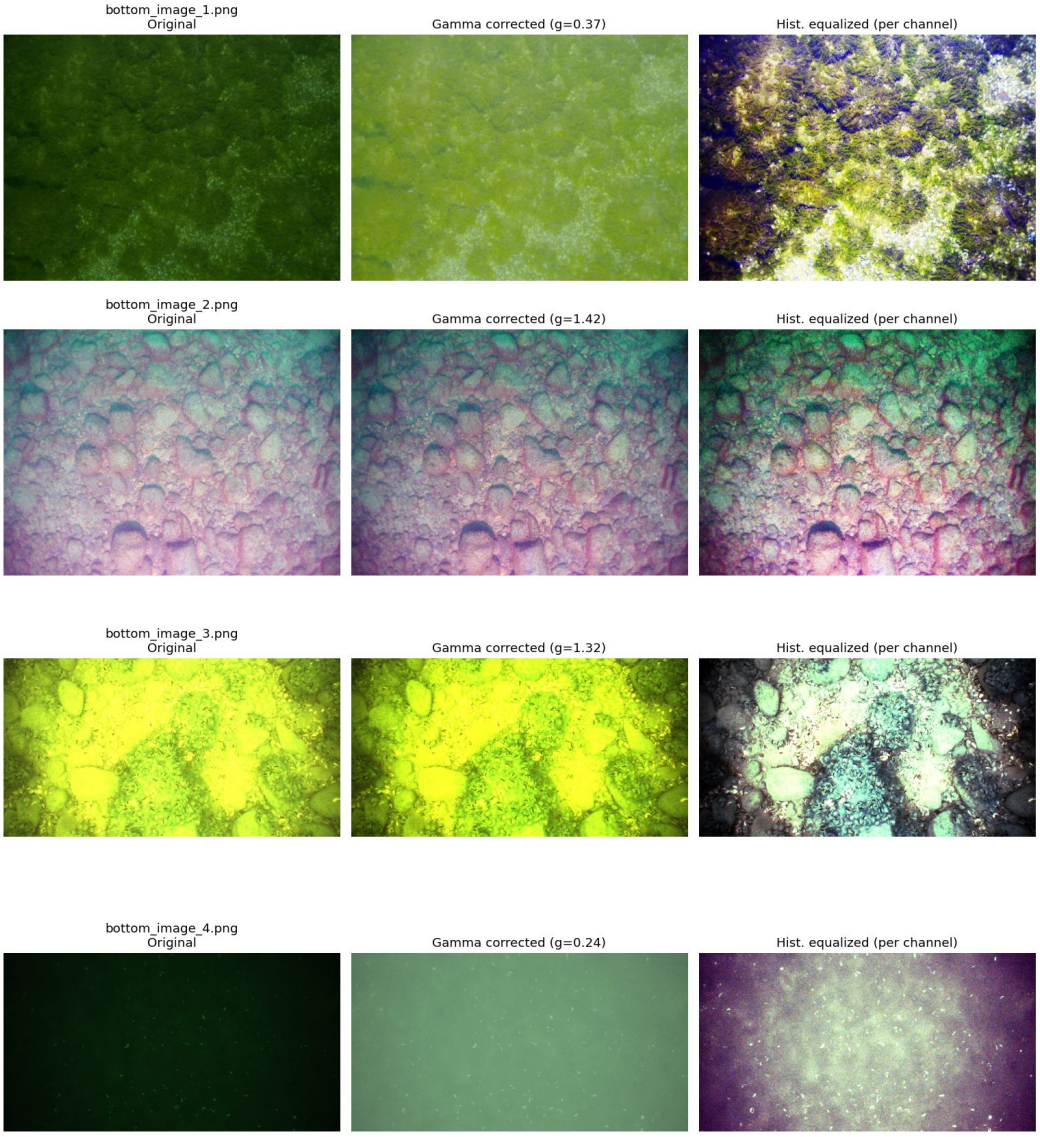


Figure 6. Original, gamma-corrected, and histogram-equalized (per RGB channel) results for each image. Gamma values were computed automatically to bring each image's mean brightness to mid-gray (128/255): Image 1 = 0.37, Image 2 = 1.42, Image 3 = 1.32, Image 4 = 0.24.

Gamma correction successfully brightened the underexposed images (1 and 4) and slightly darkened the overexposed ones (2 and 3), matching the direction expected from their histograms. However, gamma correction cannot recover detail that was already clipped: Image 3's blown-out highlights remain blown out after gamma correction, since that information was lost at capture, not just compressed into a narrow range. Histogram equalization went further, revealing considerably more texture and detail in all four images (most noticeably in Images 3 and 4), but per-channel RGB equalization introduced false-color artifacts (visible as blue-cyan patches in Image 1 and magenta patches in Image 4), a direct result of stretching a channel that was mostly noise (the crushed blue channel) into a wide, visually prominent

range.

LAB color space equalization

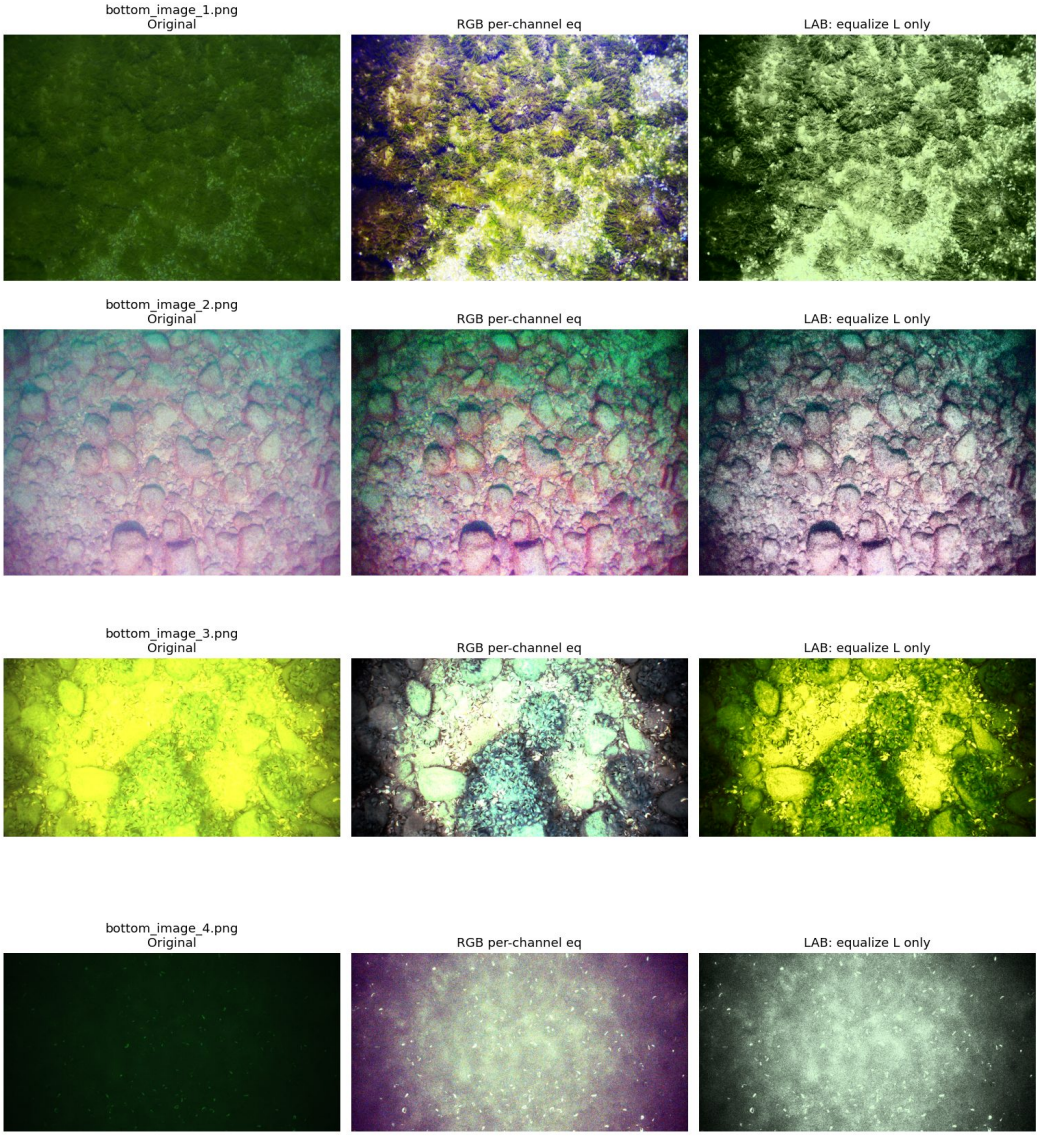


Figure 7. Comparison of RGB per-channel equalization against equalizing only the Lightness (L) channel in CIELAB space, leaving the color (a/b) channels untouched.

Equalizing only the L channel in LAB space avoided the false-color artifacts seen with RGB per-channel equalization across all four images. Because LAB separates lightness from color, boosting contrast in L does not distort the color channels the way independently stretching R, G, and B does. The trade-off is that LAB equalization does not correct the underlying color cast itself; the images remain green, magenta, or yellow-tinted, and it only improves contrast and revealed detail within that existing cast. A combined approach (a mild per-channel gamma adjustment to address the cast, followed by LAB L-channel equalization for contrast) was used going forward as the basis for smoothing and sharpening.

Smoothing, noise reduction and edge enhancement

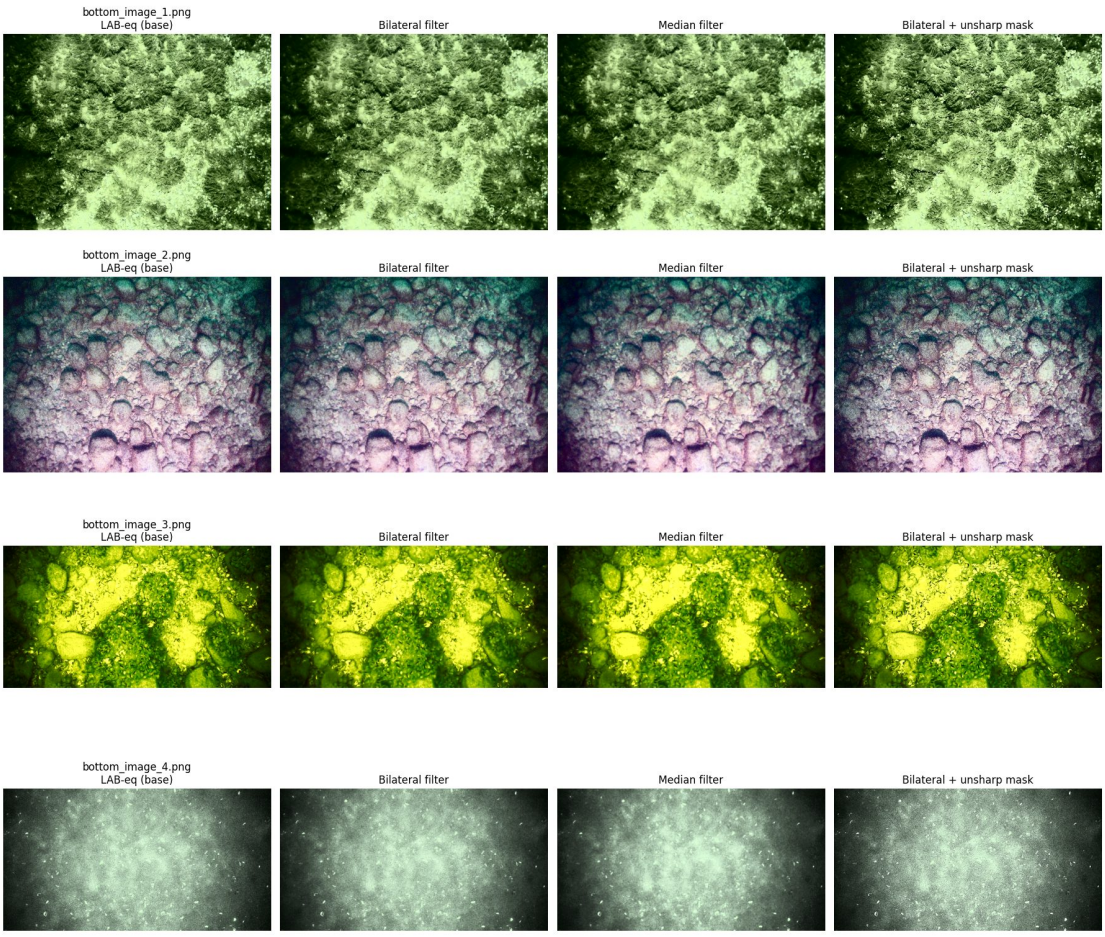


Figure 8. LAB-equalized base image compared against a bilateral filter, a median filter, and a bilateral filter followed by an unsharp mask.

Starting from the LAB-equalized result, a bilateral filter smoothed fine grainy texture noise while preserving sharp edges (most visible in Images 1 and 4). A median filter was particularly effective on Image 4, where it visibly reduced the small bright particle specks caused by backscatter, since median filtering removes isolated-pixel noise without blurring edges the way a Gaussian filter would. Applying an unsharp mask after the bilateral filter restored edge crispness that had been softened, without reintroducing the fine speckle noise; this combination produced the clearest final result across all four images.

Enhancing a new test image

The final pipeline (LAB Lightness equalization, followed by a median filter to remove particle/backscatter noise, a bilateral filter to smooth texture while preserving edges, and an unsharp mask to restore edge crispness) was applied to a separate, very low-contrast test image.

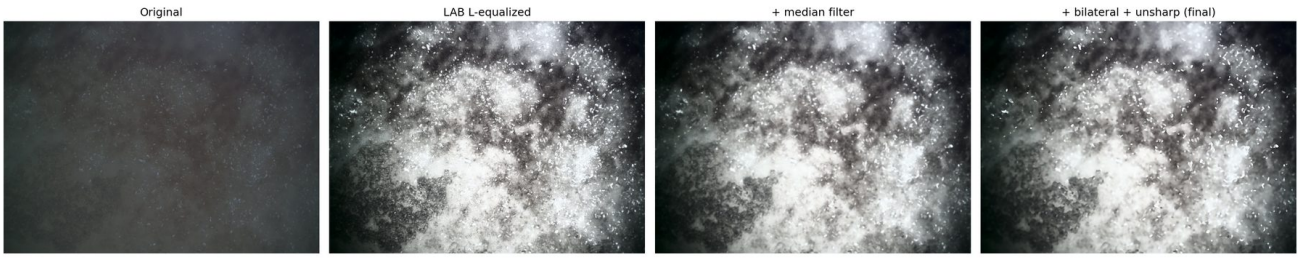


Figure 9. Test image pipeline: original, LAB L-equalized, plus median filter, and the final result after bilateral filtering and unsharp masking.



Figure 10. Final enhanced test image.

The original test image was extremely low-contrast and hazy, with almost no visible structure. LAB equalization alone revealed the vast majority of the improvement, uncovering particulate matter and surface texture that were essentially invisible in the original. The median and bilateral filtering steps then cleaned up noise introduced or revealed by that contrast boost, and the unsharp mask restored a crisp, well-defined final image.

4. What Worked, and Why

One consistent pipeline for all four images

The same processing sequence (gamma correction to normalize brightness, LAB Lightness-channel equalization for contrast, a median filter to remove particle/backscatter noise, a bilateral filter to smooth texture while preserving edges, and an unsharp mask to restore edge crispness) improved all four images. The only value that had to change per image was the gamma exponent itself, which was computed automatically from each image's mean brightness (0.37, 1.42, 1.32, and 0.24 for Images 1 through 4) rather than picked by hand, so the pipeline as a whole stayed consistent even though its one tunable parameter adapted to each image.

Which operations improved visibility, color and noise

LAB Lightness equalization gave the biggest single improvement in visibility, revealing texture and detail that was invisible in every original image, without disturbing color the way RGB per-channel equalization

did. Gamma correction improved overall visibility for the two under-exposed images (1 and 4) and slightly toned down the two over-exposed ones (2 and 3). The median filter reduced small bright noise specks from backscatter, most noticeably in Image 4, and the bilateral filter reduced grainy texture noise generally while keeping edges intact. The unsharp mask improved perceived detail and edge visibility after the smoothing steps had softened them slightly.

Operations that only worked sometimes

Per-channel RGB histogram equalization was the least consistent operation. It worked reasonably on Image 2, which had a fairly balanced starting histogram, but produced obvious false-color artifacts (blue-cyan blotches in Image 1, magenta blotches in Image 4) wherever a channel was almost entirely crushed near zero. Stretching a channel that is mostly noise, rather than signal, simply amplifies that noise into a visually prominent, unrealistic color, so this operation's success depended directly on how much real signal was left in each channel to begin with. Gamma correction was similarly limited by exposure: it could redistribute existing values but could not recover the highlights that were already clipped to white in Image 3.

Operations that always helped

LAB Lightness-channel equalization helped in every image, since it only affects contrast, not color, so it cannot introduce the false-color problem RGB equalization did. The bilateral filter and the unsharp mask also consistently helped across all four images: the bilateral filter never visibly damaged real detail while it reduced noise, and the unsharp mask reliably restored the sharpness that filtering softened, without reintroducing speckle noise.

Making the method robust to new scenes

Some of it did require scene-specific judgment: deciding by eye that Image 1's blue channel was crushed, or that Image 3's highlights were clipped, came from actually looking at each histogram. But parts of the pipeline can be made scene-agnostic: the auto-gamma step already picks its exponent automatically from each image's own mean brightness rather than a fixed value, and choosing LAB Lightness equalization instead of RGB per-channel equalization is itself a generally safe default, since it improves contrast without needing to know which specific channel is degraded. A fully automatic pipeline could extend this by checking each channel's histogram spread or skew and only equalizing channels that are unusually narrow or clipped, rather than assuming every image needs the same fixed treatment.